# A Comparative Analysis of Large Language Model(LLM) and Human-Generated Text Using Natural Language Processing

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## Introduction

### Key question: Can we still tell apart human and machine-generated text?

- This study explores "Differences" between human responses and those from major LLMs
- Modern LLMs(e.g., GPT, Gemini) produce text nearly indistinguishable from human writing
- Focused on outputs from Claude, DeepSeek, Gemini, GPT and Human
- Aims to identify "syntactic and lexical Fingerprints" for each source
- Investigates whether yet classifier works to discriminate the human despite fluent coherent writing of LLM

## Data & Methodology

- First, for the purpose of analyzing and classifying responses as either LLM-generated or human-written, we collected real-world answers from human users on Quora and generated responses from four major LLMs: Claude, Gemini, DeepSeek and GPT
- Second, we used five distinct prompt modes for varied LLM response characteristics
- We explored various linguistic features (e.g., syntactic stats, POS tags, TF-IDF, SBERT, LDA) to analyze the responses.
- For classification, we trained a LightGBM model using input features based on LDA-based subtopic distributions.

### Prompt Typology

**Statistical Analysis**

**Minimal Guidance**
- Uses minimal, neutral instructions to capture the LLM's default response style

**Objective Response**
- Directs the LLM to prioritize factual, objective information, testing its ability

**Experience Sharing**
- Aims to elicit subjective, experience-based narratives and opinions, assessing the LLM's capacity for a personal

**Analytical Response**
- Encourages detailed, analytical answers that push the LLM to explore topics comprehensively

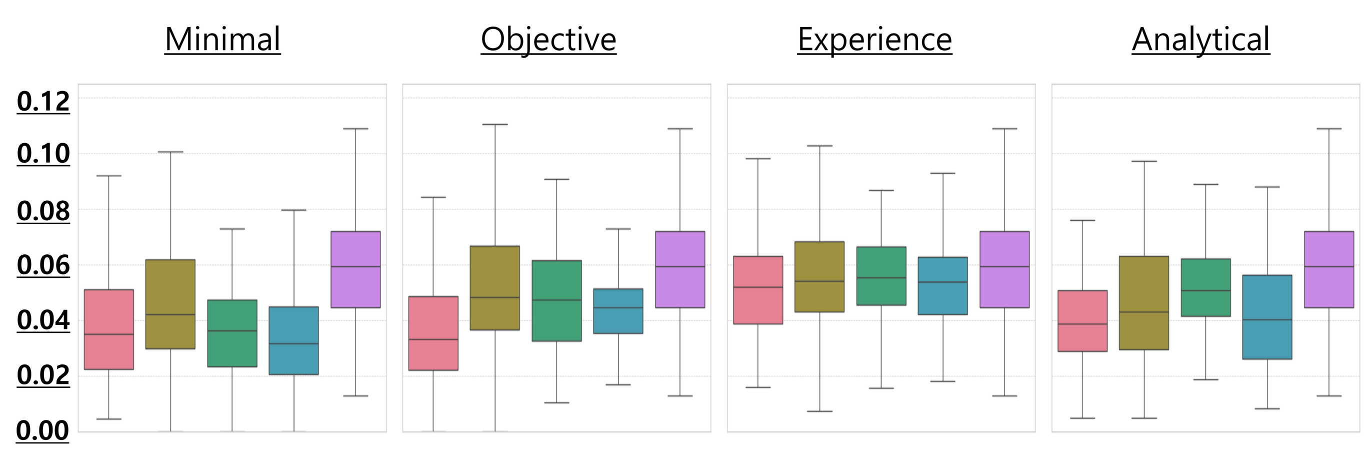
**TF-IDF, SBERT, Classfication**

**Core Informative & Analytical Set**
- Assesses versatile LLM responses by combining Minimal, Analytical, objective.
- Subjective experience prompts were excluded to first establish a standardized base-Line for core information processing capabilities.
- A total of 100 questions were curated across multiple topics, with 30 responses per question sourced from four LLMs and human contributors. While LLM responses were consistently sampled to provide 30 answers per question, the number of human responses varied, and some questions had fewer than 30 human answers available.

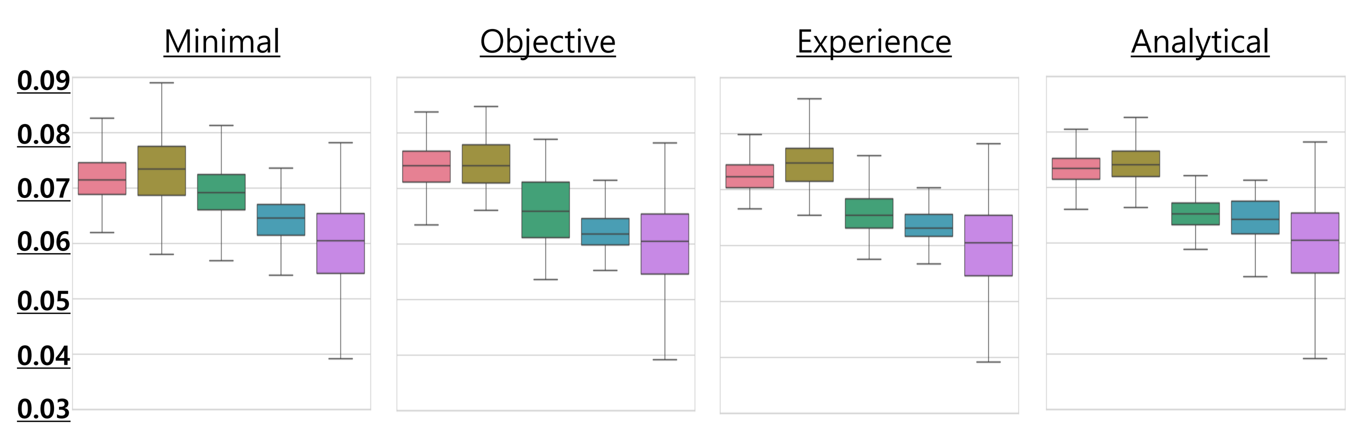
- Lastly, the input texts were cleaned by removing special characters, lowercased, tokenized using NLTK(Natural Language ToolKit), and filtered to exclude English stopwords for data preprocessing & tokenization
- Linguistic analysis(stylometry, POS) clearly distinguishes human text from LLMs. Humans show vast stylistic diversity. LLMs exhibit more consistent, model-specific signatures.

## Syntactic Fingerprinting : How They Speak

### Personal Pronoun Proportion




### Lexical Density




Claude DeepSeek Gemini GPT Human

$$Lexical\ Density = \frac{Number\ of\ Unique\ Words}{Total\ Number\ of\ Words(Tokens)}$$

$$PRP\ \ Proportion = \frac{Number\ of\ Personal\ Pronouns}{Total\ Number\ of\ Words}$$

## Topical & Lexical Signatures : What They Speak about

### Term Frequency - Inverse Document Frequency

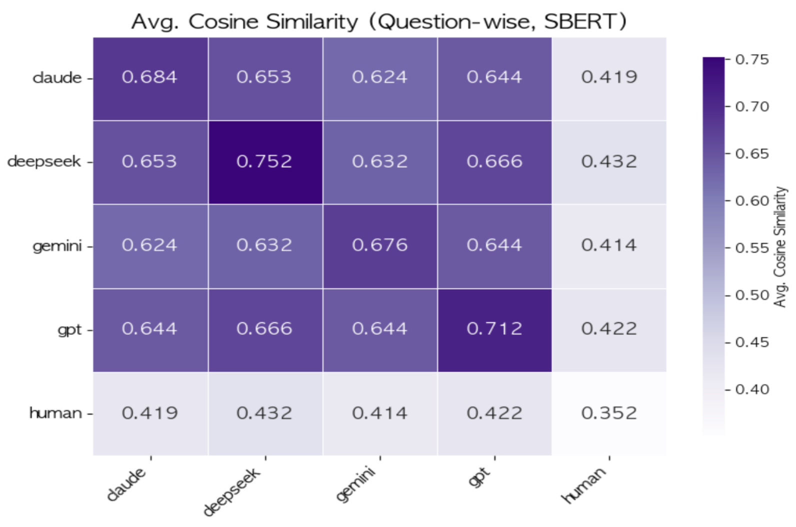
- LLMs generate responses using a common, general-purpose lexicon. In contrast, human lexical choices indicate responses more grounded in specific contexts, personal perspectives, temporal awareness

| Top-10 | Human | | GPT | | Gemini | | DeepSeek | | Claude | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Word | Score | Word | Score | Word | Score | Word | Score | Word | Score |
| 1 | One | 0.2570 | Like | 0.3072 | Like | 0.3796 | Like | 0.3088 | Like | 0.2530 |
| 2 | Time | 0.2424 | One | 0.2373 | Think | 0.2711 | One | 0.2925 | People | 0.2247 |
| 3 | **Would** | 0.2235 | Time | 0.1840 | Thing | 0.2285 | People | 0.2696 | I've | 0.2189 |
| 4 | People | 0.1879 | People | 0.1837 | One | 0.1966 | Even | 0.2426 | Think | 0.2042 |
| 5 | **Like** | 0.1807 | Think | 0.1768 | People | 0.1805 | Way | 0.1815 | Something | 0.1804 |
| 6 | Get | 0.1582 | Might | 0.1735 | Even | 0.1754 | Thing | 0.1621 | One | 0.1750 |
| 7 | Day | 0.1550 | Thing | 0.1595 | Okay | 0.1751 | Time | 0.1506 | Actually | 0.1681 |
| 8 | Year | 0.1521 | I've | 0.1563 | Really | 0.1610 | Make | 0.1402 | Thing | 0.1529 |
| 9 | *Life* | 0.1457 | Even | 0.1517 | Know | 0.1607 | Something | 0.1389 | Time | 0.1517 |
| 10 | dont | 0.1381 | Life | 0.1473 | something | 0.1361 | dont | 0.1331 | someone | 0.1344 |

- $TF - IDF = TF \ * \ IDF$
  - TF(Term Frequency) : How often a word appears in a document
  - IDF(Iverse Document Frequency) : How rare a word is across all documents

### Sentence BERT

- SBERT (sentence embedding) analysis shows LLMs are highly similar and consistent in their semantic meaning.
- Human responses are distinctly different from LLMs and far more internally diverse




### Hierarchical LDA-Based Subtopic Modeling

1. Topic Modeling (LDA)
   - Construct Dictionary and Bag-of-Words from tokenized data
   - The number of topics was set to five (four LLMs and one human)
   - Represent each document as a distribution over main topics
2. Subtopic Modeling
   - Select top 20 representative documents for each main topic
   - Identify optimal number of subtopics within each document group
   - Apply subtopic model to all documents to compute subtopic probability distributions

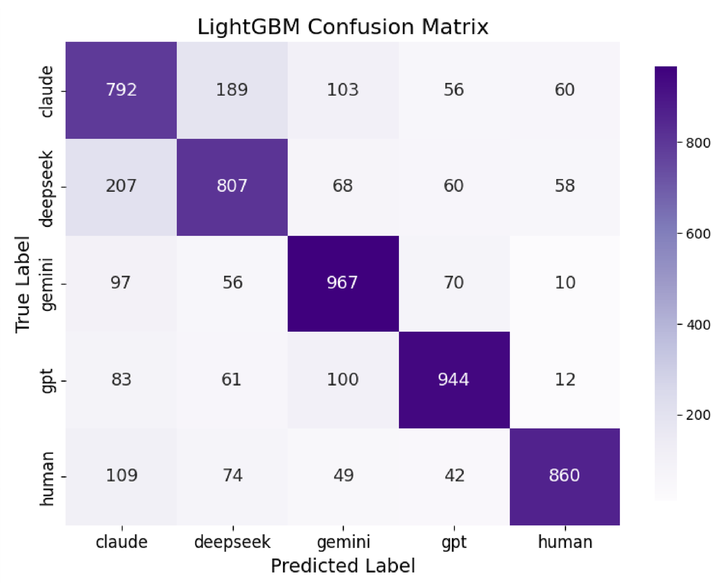
| Doc_id | 0-1 | 0-2 | 1-1 | 1-2 | 1-3 | 1-4 | ... | LLM_Name |
|---|---|---|---|---|---|---|---|---|
| 1 | 0.7245 | 0.2755 | 0.1853 | 0.2493 | 0.2679 | 0.2975 | ... | GPT-4 |
| 2 | 0.9074 | 0.0926 | 0.2296 | 0.2947 | 0.1642 | 0.3115 | ... | Claude |
| 3 | 0.8692 | 0.1308 | 0.2779 | 0.2186 | 0.2379 | 0.2656 | ... | Claude |
| 4 | 0.8739 | 0.1261 | 0.1452 | 0.2068 | 0.3081 | 0.3399 | ... | Deepseek |
| . . . . . . | | | | | | | | |
| n-1 | 0.8486 | 0.1514 | 0.2454 | 0.2154 | 0.3099 | 0.2292 | ... | Human |
| n | 0.8790 | 0.1210 | 0.1651 | 0.2923 | 0.2491 | 0.2935 | ... | Human |

* This matrix is used as the input feature set for classification

### LightGBM Based Classfication

- Input X : Subtopic probability vector per document
- Target Y : LLM name associated with each document
- Classifier : LightGBM multi-class model (Spliting Train 60% and Test 40%)

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| claude | 0.614907 | 0.66 | 0.636656 | 1200 |
| deepseek | 0.679865 | 0.6725 | 0.676163 | 1200 |
| gemini | 0.75136 | 0.805833 | 0.777644 | 1200 |
| gpt | 0.805461 | 0.786667 | 0.795953 | 1200 |
| human | 0.86 | 0.758377 | 0.805998 | 1134 |
| accuracy | 0.736434 | | | |
| Macro avg | 0.742319 | 0.736675 | 0.738483 | 5934 |
| Weighted avg | 0.74101 | 0.736434 | 0.737732 | 5934 |




- The LightGBM model had an overall accuracy of 73.6%, classifying human responses most accurately, while Claude and DeepSeek were often confused. In particular, Gemini and GPT had high f1-scores, showing that they reflect the differences in responses between LLMs.

## Conclusion & Future Work

- This study demonstrates that distinctive linguistic fingerprints exist, enabling reliable classification between human and LLM-generated text, as well as among different LLMs. These differences stem from core features like semantic diversity and narrative style.
- The classifier developed here can help combat Model Collapse, the degradation of model trained on AI generated content. By assigning an "AI-generated probability score" to training data, it can assist in filtering such content, preserving data diversity and ensuring more sustainable LLM development.

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